



frozen



trainable

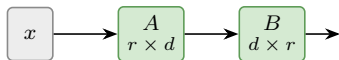


input-conditioned gate

STATIC

LoRA

$$\Delta W = BA$$

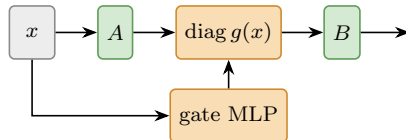


$$\text{rank}(\Delta W) \leq r \text{ for every input}$$

INPUT-GATED

Gated low-rank

$$\Delta W(x) = B \text{diag}(g(x)) A$$



$$\text{rank} = k(x) \leq r: \text{one unit of rank per gate unit}$$

KronLoRA

$$\Delta W = A \otimes (B_1 B_2)$$

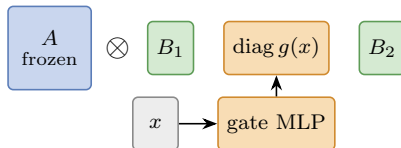


$$\text{rank}(\Delta W) = pr$$

A adds rank at zero trainable cost

DynKronLoRA

$$\Delta W(x) = A \otimes (B_1 \text{diag}(g(x)) B_2)$$



$$\text{rank} = p k(x): p \text{ units of rank per gate unit}$$

Low-rank substrate

Kronecker substrate